ReadMe Document

This ReadMe document will help you with a quick navigation through the contents of the Pre-Reading material, Preview videos, Individual Labs and Mini-Hackathon/Hackathon for the upcoming session.

Outcomes

After completion of this week's sessions, & activity participants will be able to :

1. Understand the concepts of NLP and dealing with textual data
2. Understand & implement Word Embedding and Word2Vec
3. Understand and implement the concepts of Perceptron, Multi-Layer perceptron, and Neural Networks.
4. Understand the concept of Gradient Descent and its application.

Lecture Topics

1. Word Embedding & Word2Vec
2. Intro to NLP
3. Perceptrons
4. MLP
5. Neural Networks
6. Gradient Descent

Preview Videos

The Preview Videos Link for the respective sessions are available a week in advance (i.e. every Monday), to help you understand the concepts that will be covered in the upcoming lectures. There could be one or more videos depending on the lecture details. The video/videos should be viewed before attending the lecture for better understanding.

There are 5 preview videos for this session and they should be viewed as given below:

1. Contextual Word Representation-Word2Vec:This is a 11-minute 40 sec video. This video helps you to understand the basic concept of contextual word representation.
2. Word Embedding: This is a 4-minutes 44 sec video. This video helps you to understand the concept of word representation using 1-hot encoding and how to calculate the cosine distance or euclidean distance between the words.The concept of dense word vector representation is introduced
3. NLP: This is a 2 minutes 50 sec video. This video covers the basics of NLP, NLP tasks and Tools
4. Neural Network 1: This is a 3-minutes 14 sec video. This video talks about how nonlinear classification is implemented through neural networks and helps you to understand xor problem.

5. Neural Network 2: This is a 7-minutes 33 sec video.This video will help you with a better understanding of Single Layer Perceptron, Multi Layer Perceptron and Activation Functions
6. Gradient Descent 1: This is a 2 minutes 37 sec video. In this video you will understand what y = f(x) and y = wx .
7. Gradient Descent 2: This is a 6-minutes 29 sec video. This video gives you an overview of gradient descent and how to minimize the prediction error and update W value. To minimize the error learning rate or eta is important.

Pre Reading Material

The pre-reading material is given on Word2Vec, Perceptron, Gradient Descent, MLP. You should read the material before you attend the lecture. This pre-reading will provide the gist of the lecture and facilitate your understanding during the lecture.

Reference Material

The Reference Material is given on Word2Vec, Perceptron, Gradient Descent. You should read the material after attending the lecture. This reference material will provide further depth into the topic.

Individual Labs

1. Word2Vec & NLP
2. Perceptrons
3. Gradient Descent
4. MLP